

Bacterial Choline-O-Sulfate Uptake and Desulfation: A Reusable Transport-Plus-Desulfation Module

A review-style synthesis of the two-step system in which choline-O-sulfate is imported and then hydrolyzed by the BetC choline sulfatase to choline, sulfate, and a proton.

1. Executive Summary

Choline-O-sulfate (COS) is a widespread trimethylammonium sulfate ester used by many bacteria (and fungi and archaea) as a source of osmoprotectant precursor, carbon, nitrogen, and — importantly — sulfur. This review treats the **transport-plus-desulfation boundary**: the minimal, reusable module in which extracellular COS is first moved across the cytoplasmic membrane and then cleaved by a **BetC choline sulfatase** into choline, inorganic sulfate, and a proton. Everything downstream (oxidation of choline to glycine betaine, transcriptional regulation, and sulfur-starvation regulons) lies outside the module boundary and is deliberately treated as neighboring biology.

The central conclusion of this synthesis is that the module is **genetically and mechanistically separable into an interchangeable importer and a conserved catalytic core**. Import obligatorily precedes hydrolysis — BetC is a soluble cytoplasmic enzyme, and *betC* deletion mutants still accumulate intact COS but cannot catabolize it — but the transporter that performs import is *not* molecularly fixed. It can be a dedicated BetD (substrate-binding protein) plus BetE (permease) ABC cassette, as characterized in *Pseudomonas* sp. ATCC19151, or it can be a promiscuous general osmoprotectant ABC importer (ProU/OpuC/YehZ-type) that recognizes COS as a "betaine" analog. In contrast, **BetC is the conserved defining part of the module**: a formylglycine-dependent Class I sulfatase of the arylsulfatase/alkaline-phosphatase (AP) superfamily whose crystal structure (the *Sinorhizobium meliloti* enzyme, SmCS) shows that, remarkably for an alkyl sulfate, it hydrolyzes the substrate by attack at sulfur (S–O bond cleavage) via a sulfated-formylglycine covalent intermediate.

The same chemistry is deployed to two different physiological ends depending on lineage and genomic context. In Rhizobiaceae, *betC* sits in the *betICBA* operon, feeding choline into glycine betaine synthesis for osmoprotection and full C/N/S nutrition. In a broad range of free-living microbes, *betC* occurs in a minimal cassette with just a choline-sulfate transporter and appears dedicated to fulfilling elemental sulfur requirements — a role that in *Pseudomonas putida* is explicitly uncoupled from osmoprotection. The best-supported open question is the molecular identity, structure, and true in vivo flux of the COS importer(s), since the "BetDE-style" transporter is defined largely by genomic adjacency and can evidently be substituted by promiscuous betaine transporters.

2. Definition and Biological Boundaries

What the system is

The module comprises exactly two catalytic/transport steps:

1. **COS import** — translocation of choline-*O*-sulfate from the periplasm/environment into the cytoplasm, carried out by a **BetDE-style** dedicated ABC transporter (substrate-binding protein BetD + permease BetE) *or* by a promiscuous betaine/osmoprotectant ABC importer.
2. **COS hydrolysis** — cytoplasmic cleavage of the C–O–SO₃[−] ester by **BetC**, a choline sulfatase, yielding choline + sulfate + H⁺.

The defining relationship is directional and obligatory: **transport supplies cytoplasmic COS for BetC hydrolysis**. Because BetC is soluble and cytoplasmic, hydrolysis cannot occur until import has taken place. This is not merely inferred — it is demonstrated by the *P. putida betC* mutant, which accumulates intact COS internally but cannot use it as a carbon or nitrogen source ([PMID: 17116241](#)).

What is adjacent but out of scope

Several neighboring processes are frequently discussed together with COS metabolism and are easily conflated with the module, but should be treated separately:

- **Choline oxidation to glycine betaine.** The BetA (choline dehydrogenase) and BetB (betaine aldehyde dehydrogenase) enzymes convert the *product* of the module (choline) into the osmoprotectant glycine betaine. These are downstream and are not part of the transport-

plus-desulfation boundary, even though in Rhizobiaceae they are encoded in the same *betICBA* operon.

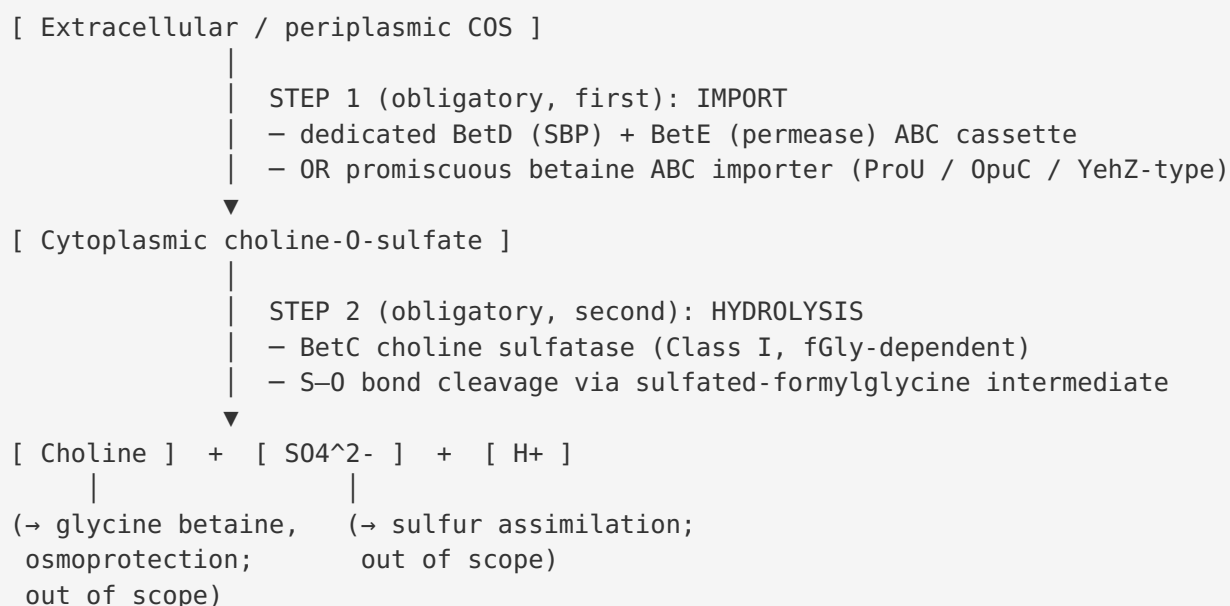
- **Transcriptional regulation.** Regulators such as BetI (a choline-responsive repressor in the *betICBA* operon) and BetR (a LysR-type activator of *betC* in *Pseudomonas* sp. ATCC19151) control *when* the module is expressed but are not themselves part of the catalytic path.
- **General osmoprotectant transport and biosynthesis.** ProU, OpuC, OpuD, OusB, BetU, ProP, and related systems import many trimethylammonium compounds. They intersect the module only insofar as some of them can also import COS; their broader osmoregulatory roles are out of scope.
- **Alternative sulfur-starvation pathways.** COS desulfation is one of many routes bacteria use to acquire sulfur under sulfate limitation; other organosulfur scavenging systems are neighboring but distinct.

Competing definitions in the literature

The chief definitional tension is **whether "the importer" is a specific pair of proteins or a functional role**. The name "BetDE" derives from the *Pseudomonas* sp. ATCC19151 cassette, where *betD* encodes a PBPB-superfamily substrate-binding protein and *betE* a STAS-domain-containing permease ([PMID: 21369825](#)). But equilibrium-dialysis and ITC data show COS is a genuine ligand of the general osmoprotectant binding proteins ProX and YehZ ([PMID: 26325238](#)). Thus one can define the import step narrowly (a dedicated BetDE cassette) or functionally (any transporter that delivers cytoplasmic COS). This review adopts the functional definition and treats BetDE as the prototypical, but not obligatory, importer.

3. Mechanistic Overview

Best current model of the sequence of events



Obligatory steps. Both import and hydrolysis are obligatory for productive COS utilization, and they must occur in this order because BetC is cytoplasmic. There is no evidence for periplasmic or extracytoplasmic COS desulfation in these systems.

Conditional / accessory features. The *identity* of the importer is conditional on genomic context (dedicated cassette vs. promiscuous transporter). Enzyme maturation — the post-translational conversion of the BetC active-site cysteine (or serine) to Ca-formylglycine — is an accessory but essential prerequisite: without it, BetC has no catalytic residue. This maturation is performed by formylglycine-generating machinery (aerobic FGE, or the anaerobic radical-SAM anSME), and is a hidden dependency of the whole module.

The catalytic chemistry of BetC

BetC belongs to the **group I (formylglycine-dependent) sulfatases**. Its catalytic residue is a Ca-formylglycine (fGly) generated post-translationally from a cysteine embedded in the diagnostic **C-X-P-X-R** sulfatase signature ([PMID: 21369825](#)). The fGly aldehyde is hydrated to a geminal diol; one hydroxyl attacks the sulfur of the substrate's sulfate ester, forming a **covalent sulfated-formylglycine intermediate**, which then eliminates sulfate to regenerate the aldehyde ([PMID: 25514000](#); [PMID: 24555731](#)).

The mechanistically striking point, established by the SmCS crystal structure, is that COS is an **alkyl sulfate**, and alkyl sulfatases normally cleave the **C–O** bond (nucleophilic attack at carbon). BetC/SmCS instead behaves like an **arylsulfatase**, cleaving the **S–O** bond (attack at sulfur), despite the alkyl nature of its substrate ([PMID: 29458126](#)). This makes BetC a chemically noteworthy Class I sulfatase specialized for an alkyl sulfate ester.

4. Major Molecular Players and Active Assemblies

Step	Player	Family / fold	Role / activity	Key evidence
Import (dedicated)	BetD	PBPb superfamily substrate-binding protein	Periplasmic COS binding for ABC import	PMID: 21369825
Import (dedicated)	BetE	Permease with STAS domain	Membrane translocation of COS	PMID: 21369825
Import (promiscuous)	ProX / ProU, YehZ, OpuC, OusB	Betaine/osmoprotectant ABC importers	Recognize COS as a betaine analog	PMID: 26325238 ; PMID: 16000740
Hydrolysis	BetC	Class I fGly-dependent sulfatase; SulfAtlas family S1; AP superfamily	Choline-O-sulfate → choline + sulfate + H ⁺ (S–O cleavage)	PMID: 9736747 ; PMID: 29458126
Maturation (accessory)	FGE / anSME	Cu-dependent FGE (aerobic); SPASM-domain radical-SAM (anaerobic)	Cys/Ser → Cα-formylglycine conversion	PMID: 23650368

The importer: interchangeable, not fixed

Two lines of evidence establish that the import step is served by more than one kind of transporter. First, the dedicated cassette: in *Pseudomonas* sp. ATCC19151 the *bet* locus encodes BetC (choline sulfatase), BetD (SBP), BetE (permease), and BetR (divergent LysR-type regulator

that activates *betC* in the presence of choline sulfate) (PMID: 21369825). Second, the promiscuous route: choline-*O*-sulfate binds the periplasmic binding proteins of *E. coli* general osmoprotectant ABC transporters — ProX (ProU system) with micromolar affinity and YehZ with millimolar affinity — alongside glycine betaine and DMSP (PMID: 26325238). Broad-specificity betaine/choline importers such as OusB from *Erwinia chrysanthemi* transport structurally related trimethylammonium osmolytes (PMID: 16000740). Consequently, COS can enter the cytoplasm without a dedicated BetDE system.

BetC: the conserved core

BetC was first defined genetically in *S. meliloti*, where *betC* encodes a choline sulfatase that converts COS (and, more slowly, phosphorylcholine) to choline; activity is absent in *betC* mutants (PMID: 9736747). Its placement in the alkaline-phosphatase/arylsulfatase (alkPPc) superfamily and possession of the CXPXR signature mark it as a group I cysteine-type sulfatase (PMID: 21369825). The authentic enzyme's structure and mechanism (SmCS) confirm it is a Class I sulfatase efficient on the alkyl sulfate COS (PMID: 29458126).

5. Evolutionary and Cell-Biological Variation

Deep origin: an ancient AP-superfamily scaffold

BetC sits within the **formylglycine-dependent sulfatases**, which SulfAtlas classifies as the **largest of the four sulfatase families**, subdivided by phylogeny into 73 subfamilies, each mapping to a substrate specificity (PMID: 27749924); choline sulfatases form one such alkyl-sulfate-specific subfamily. This family lies within the **alkaline-phosphatase (AP) superfamily**, whose members share a bimetallo active-site scaffold and pronounced **catalytic promiscuity** — interconverting arylsulfatase, phosphomonoesterase, and phosphodiesterase activities. This promiscuity is widely argued to be the evolutionary "springboard" from which new specificities such as BetC's arose (PMID: 30497259; PMID: 29070681). The AP scaffold is therefore the **deepest conserved element** of the module, far older than COS metabolism itself, while the alkyl-sulfate specificity of choline sulfatases is a **later, subfamily-level elaboration**.

The maturation machinery is likewise ancient and split by oxygen dependence: aerobic organisms use the copper-dependent FGE, while anaerobes use a SPASM-domain radical-SAM anaerobic sulfatase-maturing enzyme (anSME) that generates fGly using AdoMet radical chemistry (PMID: 23650368).

Two functional configurations of *betC*

A genomic survey resolves the module into two recurring architectures with distinct physiological meanings (PMID: 24281732):

Feature	Osmoprotection / nutrition	Sulfur scavenging
Genomic context	<i>betICBA</i> operon	Minimal cassette: <i>betC</i> + choline-sulfate transporter
Taxonomic distribution	Restricted to Rhizobiaceae	Wide: free-living bacteria, archaea, fungi
Downstream fate of choline	Oxidized to glycine betaine (osmoprotectant)	Choline released; sulfate is the prize
Physiological trigger	Choline/osmotic context; C/N/S nutrition	Elemental sulfur requirement
Discriminating marker	—	A signature sequence separates the two functions

In *S. meliloti*, COS is metabolized to the osmoprotectant glycine betaine and can serve as sole C, N, or S source (PMID: 9736747). In *P. putida*, by contrast, *betC* is functionally **uncoupled from osmoprotection**: *betC* expression is *downregulated* at high salt, showing that its principal role is COS metabolism (sulfur/carbon acquisition) rather than compatible-solute production (PMID: 17116241). This is the clearest evidence that the same two-step chemistry has been recruited independently for different ends across lineages.

Alternative routes to the same outcome

The most important "alternative route" is at the **import step**: dedicated BetDE cassette versus promiscuous betaine ABC importers achieve the identical outcome (cytoplasmic COS delivery) by different molecular means. At the **hydrolysis step**, no non-BetC route to COS desulfation is established in these systems; BetC (or a close choline-sulfatase homolog) appears to be the sole catalytic solution recorded here.

6. Constraints, Dependencies, and Failure Modes

Ordering constraint (compartmental). Import must precede hydrolysis. BetC is a soluble cytoplasmic enzyme, so COS must physically reach the cytoplasm before it can be cleaved. The *P. putida betC* mutant provides the decisive evidence: it accumulates intact internal COS but cannot use it as C or N source, proving that (i) import is BetC-independent and (ii) catabolism strictly requires cytoplasmic BetC (PMID: 17116241). This experiment rules out the otherwise plausible model of periplasmic desulfation followed by choline uptake.

Maturation dependency. BetC is catalytically dead without post-translational conversion of its active-site Cys/Ser to formylglycine. The module therefore has a hidden dependency on FGE (aerobic) or anSME (anaerobic) maturation systems (PMID: 23650368; PMID: 25514000). A cell lacking functional maturation machinery would import COS but fail to desulfate it — a failure mode phenotypically resembling a *betC* null.

Substrate/mechanism constraint. COS is an alkyl sulfate, and the "default" chemistry for such substrates is C–O cleavage. BetC instead enforces S–O cleavage via the sulfated-fGly intermediate (PMID: 29458126; PMID: 24555731). This is a mechanistic constraint imposed by the fGly catalytic apparatus rather than by the substrate.

Regulatory gating (out of scope but relevant to failure modes). Whether the module is expressed depends on activators/repressors (BetR, BetI) whose loss changes the physiological output without changing the underlying chemistry.

7. Controversies and Open Questions

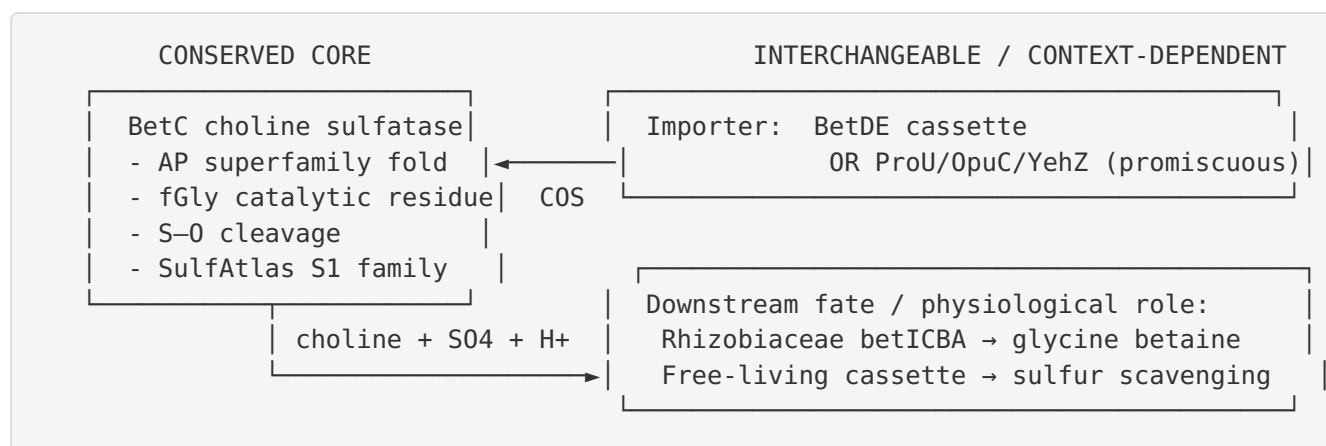
1. **What is the true COS importer, and what is its flux?** This is the single largest gap. The "BetDE-style" transporter is defined chiefly by genomic adjacency to *betC* (PMID: 21369825), yet promiscuous betaine ABC importers demonstrably bind COS (PMID: 26325238). No high-resolution structure of a dedicated COS importer, and no quantitative comparison of in vivo COS flux through dedicated versus promiscuous routes, is available. Binding (ProX micromolar, YehZ millimolar) is not the same as productive transport, so the relative physiological contributions remain unresolved.

2. **How general is the "one signature discriminates two functions" claim?** The genomic-survey distinction between the Rhizobiaceae *betICBA* operon and the free-living sulfur-scavenging cassette rests on comparative genomics plus a proposed signature sequence (PMID: 24281732). How robustly this signature predicts function across the many uncharacterized subfamilies is not experimentally validated at scale.
3. **Cross-organism extrapolation.** Much of the mechanistic detail comes from a few models — *S. meliloti* (genetics + structure), *Pseudomonas* spp. (regulation + uncoupling), and *E. coli* (transporter binding). Whether conclusions transfer to archaea and fungi that also carry *betC*-type cassettes is assumed rather than demonstrated.
4. **Maturation under anaerobiosis.** The requirement for anSME-type maturation in anaerobes that use BetC is inferred from general sulfatase biology (PMID: 23650368) rather than directly shown for choline sulfatases specifically.
5. **Substrate breadth of BetC.** BetC also acts, more slowly, on phosphorylcholine (PMID: 9736747), consistent with the promiscuity of AP-superfamily enzymes. The physiological relevance of this secondary activity is unclear.

Strongly supported claims include: BetC's identity as a Class I fGly-dependent choline sulfatase (genetics + structure + superfamily assignment); the obligatory import-before-hydrolysis ordering (mutant phenotype); and the existence of two lineage-specific functional configurations (comparative genomics + expression data). The weakest links are the molecular identity/flux of the importer and the generality of cross-lineage extrapolations.

8. Mechanistic Model / Interpretation (Synthesis)

The COS uptake-and-desulfation module is best understood as a **conserved enzymatic core (BetC) bolted onto an interchangeable import module**. Evolution has kept the chemistry constant — an ancient AP-superfamily sulfatase scaffold, matured to fGly, cleaving the S–O bond of an alkyl sulfate — while freely swapping the transport solution and repurposing the downstream fate of the product.



This "conserved-core / swappable-periphery" architecture explains three otherwise puzzling observations at once: (i) why *betC* appears in wildly different genomic neighborhoods across bacteria, archaea, and fungi; (ii) why the same reaction serves osmoprotection in one lineage and sulfur nutrition in another; and (iii) why import can proceed without any dedicated transporter. It also identifies BetC as the correct anchor for defining and detecting the module bioinformatically.

9. Limitations and Knowledge Gaps

- **Data type.** This review is a literature synthesis; no primary experimental dataset was analyzed. Conclusions inherit the limitations of the underlying studies.
- **Importer uncertainty.** As emphasized above, the dedicated COS importer is structurally uncharacterized, and the balance between dedicated and promiscuous import in vivo is unknown.
- **Model-organism concentration.** Mechanistic depth is uneven, with structural/mechanistic data essentially from a single enzyme (SmCS) and transport data from *E. coli* proteins whose COS-transport (as opposed to COS-binding) capacity is inferred.
- **Maturation not directly tied to BetC in anaerobes.** The anSME requirement is a reasonable extrapolation, not a choline-sulfatase-specific result.
- **Regulation deliberately excluded.** Because regulation is out of scope, statements about "when" the module operates are intentionally shallow.

10. Proposed Follow-up Experiments / Actions

1. **Structurally characterize a dedicated COS importer.** Solve the structure of a BetD substrate-binding protein (and, ideally, the BetDE complex) in complex with COS to define the recognition determinants that distinguish it from general betaine SBPs.
 2. **Quantify COS flux through dedicated vs. promiscuous routes.** Use isogenic strains lacking either the dedicated BetDE cassette or the general osmoprotectant importers, and measure $^{14}\text{C}/^{35}\text{S}$ -COS uptake kinetics to establish which transporter dominates under sulfur-limited vs. osmotic-stress conditions.
 3. **Test the sulfur-scavenging hypothesis directly.** In a free-living organism with the minimal *betC* + transporter cassette, assay growth on COS as sole sulfur source with and without *betC*, and confirm ^{35}S incorporation into cellular sulfur pools.
 4. **Validate the discriminating signature sequence.** Experimentally test whether the proposed sequence signature ([PMID: 24281732](#)) predicts osmoprotection vs. sulfur-scavenging function across a diverse, unbiased set of *betC* homologs, including archaeal and fungal representatives.
 5. **Confirm fGly maturation in anaerobic BetC users.** Directly demonstrate anSME-dependent maturation of a choline sulfatase in an anaerobe (e.g., by mass spectrometry of the active-site residue in wild-type vs. anSME-null backgrounds).
 6. **Probe BetC substrate breadth and mechanism.** Extend the SmCS structural work with substrate-analog complexes and Brønsted analyses to quantify the S–O vs. C–O cleavage preference on additional alkyl sulfates and on phosphorylcholine.
-

11. Key References

- **BetC as a choline sulfatase (genetics).** *Presence of a gene encoding choline sulfatase in Sinorhizobium meliloti bet operon.* [PMID: 9736747](#) — defines BetC activity and substrate; establishes COS → choline (+ slower phosphorylcholine).
- **BetDE cassette and regulation.** *Inducible expression of choline sulfatase and its regulator BetR in Pseudomonas sp. ATCC19151.* [PMID: 21369825](#) — defines *betC/betD/betE/betR*; places BetC in the alkPPc superfamily via the CXPXR signature.

- **Import-before-hydrolysis and uncoupling from osmoprotection.** *Uncoupling of choline-O-sulphate utilization from osmoprotection in Pseudomonas putida.* PMID: 17116241 — *betC* mutant accumulates intact COS; expression down at high salt.
- **Two functional configurations.** *The diversity and functions of choline sulphonatases in microorganisms.* PMID: 24281732 — Rhizobiaceae *betICBA* operon vs. free-living sulfur-scavenging cassette.
- **Structure and mechanism of authentic BetC.** *Structural and Mechanistic Analysis of the Choline Sulfatase from Sinorhizobium meliloti: A Class I Sulfatase Specific for an Alkyl Sulfate Ester.* PMID: 29458126 — S–O cleavage of an alkyl sulfate.
- **fGly catalysis.** *Formylglycine, a post-translationally generated residue with unique catalytic capabilities.* PMID: 25514000; and *Experimental and theoretical insights into sulfate ester hydrolysis.* PMID: 24555731.
- **Sulfatase classification.** *Matching the Diversity of Sulfated Biomolecules: SulfAtlas.* PMID: 27749924 — largest family, 73 subfamilies.
- **AP-superfamily promiscuity as evolutionary springboard.** PMID: 30497259; PMID: 29070681.
- **Anaerobic fGly maturation.** *X-ray structure of an AdoMet radical activase.* PMID: 23650368.
- **Promiscuous betaine importers bind COS.** *YehZYXW of E. coli is a low-affinity betaine-specific ABC transporter.* PMID: 26325238; *OusB from Erwinia chrysanthemi.* PMID: 16000740.

Prepared as a commissioned review synthesis. Claims are anchored to the cited primary literature; uncertainty — especially regarding the molecular identity and in vivo flux of the COS importer — is flagged explicitly throughout.
